APTITUDE CHEAT SHEET

PLACEMENT PREPARATION GUIDE

QUICK **REVISION** | SMART **SHORTCUTS** | SOLVE **FASTER**

COVERS 11 IMPORTANT TOPICS

1. TIME & WORK
2. PERCENTAGE
3. PROFIT & LOSS
4. RATIO & PROPORTION
5. AVERAGE
6. SQUARE NUMBERS
7. PROBABILITY
8. SIMPLE & COMPOUND INTEREST
9. NUMBERS
10. NUMBER SYSTEM
11. PIPES & CISTERNS

SWIPE TO EXPLORE →

SQUARE NUMBERS

[1 to 100]

$1^2 = 1$	$26^2 = 676$	$51^2 = 2601$	$76^2 = 5776$
$2^2 = 4$	$27^2 = 729$	$52^2 = 2704$	$77^2 = 5929$
$3^2 = 9$	$28^2 = 784$	$53^2 = 2809$	$78^2 = 6084$
$4^2 = 16$	$29^2 = 841$	$54^2 = 2916$	$79^2 = 6241$
$5^2 = 25$	$30^2 = 900$	$55^2 = 3025$	$80^2 = 6400$
$6^2 = 36$	$31^2 = 961$	$56^2 = 3136$	$81^2 = 6561$
$7^2 = 49$	$32^2 = 1024$	$57^2 = 3249$	$82^2 = 6724$
$8^2 = 64$	$33^2 = 1089$	$58^2 = 3364$	$83^2 = 6889$
$9^2 = 81$	$34^2 = 1156$	$59^2 = 3481$	$84^2 = 7056$
$10^2 = 100$	$35^2 = 1225$	$60^2 = 3600$	$85^2 = 7225$
$11^2 = 121$	$36^2 = 1296$	$61^2 = 3721$	$86^2 = 7396$
$12^2 = 144$	$37^2 = 1369$	$62^2 = 3844$	$87^2 = 7569$
$13^2 = 169$	$38^2 = 1444$	$63^2 = 3969$	$88^2 = 7744$
$14^2 = 196$	$39^2 = 1521$	$64^2 = 4096$	$89^2 = 7921$
$15^2 = 225$	$40^2 = 1600$	$65^2 = 4225$	$90^2 = 8100$
$16^2 = 256$	$41^2 = 1681$	$66^2 = 4356$	$91^2 = 8281$
$17^2 = 289$	$42^2 = 1764$	$67^2 = 4489$	$92^2 = 8464$
$18^2 = 324$	$43^2 = 1849$	$68^2 = 4624$	$93^2 = 8649$
$19^2 = 361$	$44^2 = 1936$	$69^2 = 4761$	$94^2 = 8836$
$20^2 = 400$	$45^2 = 2025$	$70^2 = 4900$	$95^2 = 9025$
$21^2 = 441$	$46^2 = 2116$	$71^2 = 5041$	$96^2 = 9216$
$22^2 = 484$	$47^2 = 2209$	$72^2 = 5184$	$97^2 = 9409$
$23^2 = 529$	$48^2 = 2304$	$73^2 = 5329$	$98^2 = 9604$
$24^2 = 576$	$49^2 = 2401$	$74^2 = 5476$	$99^2 = 9801$
$25^2 = 625$	$50^2 = 2500$	$75^2 = 5625$	$100^2 = 10000$

NUMBERS

- Natural numbers : 1, 2, 3, 4, 5, ...
- Even numbers : 2, 4, 6, 8, 10, ...
- Odd numbers : 1, 3, 5, 7, 9, ...
- Integer numbers : −3, −2, −1, 0, 1, 2, 3, ...
- Whole numbers : 0, 1, 2, 3, 4, ...
- Composite numbers : 4, 6, 8, 9, ...
- Prime numbers : 2, 3, 5, 7, 11, ...
- Co-prime numbers : (5, 7), (2, 3)
- Rational numbers : $\sqrt{4}$, $\frac{7}{5}$, $\frac{2}{3}$, 3
- Irrational numbers : $\sqrt{5}$, $\sqrt{7}$, $\sqrt{11}$, $\sqrt{13}$, π
- Real numbers : $\sqrt{4}$, $11\frac{4}{7}$
- Surreal numbers : $\sqrt{-6}$, $\sqrt{-5}$, $\sqrt{-29}$
- The smallest prime number is : 2 (two)
- The smallest even prime number is : 2 (two)
- The smallest composite number is : 4 (four)
- The smallest whole number is : 0 (zero)
- The smallest natural number is : 1 (one)
- π (pi) : Irrational number

- π (pi) is approximately : $\frac{22}{7}$ or 3.14159

PERCENTAGE CHEATSHEET

$\frac{1}{3}$ = 33.33%	$\frac{5}{6}$ = 83.33%	$\frac{7}{8}$ = 87.50%	$\frac{5}{11}$ = 45.45%
$\frac{2}{3}$ = 66.66%	$\frac{1}{7}$ = 14.28%	$\frac{1}{9}$ = 11.11%	$\frac{6}{11}$ = 54.54%
$\frac{1}{4}$ = 25%	$\frac{2}{7}$ = 28.56%	$\frac{2}{9}$ = 22.22%	$\frac{7}{11}$ = 63.63%
$\frac{2}{4}$ = 50%	$\frac{3}{7}$ = 42.84%	$\frac{3}{9}$ = 33.33%	$\frac{8}{11}$ = 72.72%
$\frac{3}{4}$ = 75%	$\frac{4}{7}$ = 57.14%	$\frac{4}{9}$ = 44.44%	$\frac{9}{11}$ = 81.81%
$\frac{1}{5}$ = 20%	$\frac{5}{7}$ = 71.42%	$\frac{5}{9}$ = 55.55%	$\frac{10}{11}$ = 90.90%
$\frac{2}{5}$ = 40%	$\frac{6}{7}$ = 85.71%	$\frac{6}{9}$ = 66.66%	$\frac{1}{12}$ = 8.33%
$\frac{3}{5}$ = 60%	$\frac{1}{8}$ = 12.5%	$\frac{7}{9}$ = 77.77%	$\frac{1}{13}$ = 7.69%
$\frac{4}{5}$ = 80%	$\frac{2}{8}$ = 25%	$\frac{8}{9}$ = 88.88%	$\frac{1}{14}$ = 7.14%
$\frac{1}{6}$ = 16.66%	$\frac{3}{8}$ = 37.50%	$\frac{1}{11}$ = 9.09%	$\frac{1}{15}$ = 6.66%
$\frac{2}{6}$ = 33.33%	$\frac{4}{8}$ = 50%	$\frac{2}{11}$ = 18.18%	$\frac{1}{16}$ = 6.25%
$\frac{3}{6}$ = 50%	$\frac{5}{8}$ = 62.50%	$\frac{3}{11}$ = 27.27%	$\frac{1}{24}$ = 4.16%
$\frac{4}{6}$ = 66.66%	$\frac{6}{8}$ = 75%	$\frac{4}{11}$ = 36.36%	$\frac{1}{40}$ = 2.50%

AVERAGE FORMULAS

1. AVERAGE $= \dfrac{\text{Sum of all observations}}{\text{Number of observations}}$

2. Average of first n natural numbers $= \dfrac{(n+1)}{2}$

3. Average of natural numbers up to n $= \dfrac{(n+1)}{2}$

4. Average of first n even natural numbers $= \dfrac{n}{2}$ — 2n

5. Average of first n odd natural numbers $= \dfrac{(n+2)}{2}$ — 2n–1

6. Average of first n odd numbers $= n$ — n

7. Average of first n even numbers $= n+1$ — n+1

8. Average of squares of first n natural numbers $= \dfrac{(n+1)(2n+1)}{6}$ — x^2

9. Average of cubes of first n natural numbers $= \dfrac{n(n+1)^2}{4}$ — x^3

10. If the number of observations is even, then
    Average $= \dfrac{\text{First middle term} + \text{Last middle term}}{2}$

11. Change in value of a quantity $=$ (New Average × New Number of Observations) − (Old Average × Old Number of Observations)

12. New Average $= \dfrac{\text{New Total Sum}}{\text{New Number of Observations}}$

TIME & WORK

1. BASIC FORMULA

$\boxed{\text{Work (W)} = \text{Rate} \times \text{Time}}$

$\boxed{\text{Rate (n)} = \dfrac{\text{Work}}{\text{Time}}}$

☞ Work = Constant (for same question)

2. LCM METHOD (MOST IMPORTANT)

(Q) A can do a work in 10 days, B in 20 days. Find time taken by A + B.

LCM of 10 & 20 = $\boxed{\text{20 units (TW)}}$

A's 1 day work = $\dfrac{20^{TW}}{10}$ = 2 units/day

B's 1 day work = $\dfrac{20^{TW}}{20}$ = 1 unit/day

A + B = 3 units/day

⏱ Time = $\dfrac{20}{3}$ days (Ans)

3. THREE PERSON CONCEPT

(Q) A = 15 days, B = 20 days, A+B+C = 8 days
Find C alone.

LCM of 15, 20, 8 = $\boxed{\text{120 units (TW)}}$

A = $\dfrac{120}{15}$ = 8 units/day

B = $\dfrac{120}{20}$ = 6 units/day

A+B+C = $\dfrac{120}{8}$ = 15 units/day

C = 15 − (8+6) = 1 unit/day

⏱ Time = $\dfrac{120}{1}$ = 120 days (Ans)

4. LEAVING MIDWAY

(Q) A = 10 days, B = 20 days
They work for 4 days, B leaves.
Find remaining time by A.

LCM = $\boxed{\text{20 units (TW)}}$

A = $\dfrac{20}{10}$ = 2 units/day

B = $\dfrac{20}{20}$ = 1 unit/day

Work in 4 days = (2+1) × 4 = 12 units

Remaining work = 20 − 12 = 8 units

A alone time = $\dfrac{8}{2}$ = 4 days (Ans)

5. MAN × DAY = WORK

$\boxed{\text{Men} \times \text{Days} = \text{Constant}}$

$\boxed{M_1D_1 = M_2D_2}$

(Q) 25 men → 12 days
30 men → ? days

$25 \times 12 = 30 \times x$

$x = \dfrac{25 \times 12}{30}$ = 10 days (Ans)

6. MEN & WOMEN PROBLEMS

(Q) 15 men + 20 women → 10 days
24 men + 32 women → ?

Let 1 man = 2 units, 1 woman = 1 unit

Team-1 work = (15×2 + 20×1) = 50 units/day

Team-2 work = (24×2 + 32×1) = 80 units/day

Total Work = 50 × 10 = 500 units

Days = $\dfrac{500}{80}$ = 6.25 days (Ans)

7. CONCEPT OF MDH (MAN × DAY × HOURS)

$\boxed{\text{Work} = M \times D \times H}$

(Q) A → 6 days, 5 hrs/day (Total hrs = 30)
B → 15 days, 3 hrs/day (Total hrs = 45)
A+B → ?

Total work = LCM(30, 45) = $\boxed{\text{90 units (TW)}}$

A = $\dfrac{90}{30}$ = 3 units/hr

B = $\dfrac{90}{45}$ = 2 units/hr

A+B = 5 units/hr

Time = 90 ÷ 5 = 18 hrs (Ans)

8. EFFICIENCY CONCEPT

If A is 60% more efficient than B

A : B = 160 : 100 = 8 : 5

Time ratio = 5 : 8

9. ALTERNATE DAYS

(Q) A = 20 days, B = 30 days
Work on alternate days starting with A.

LCM = $\boxed{\text{60 units (TW)}}$

A = 3 units/day

B = 2 units/day

1st day (A) = 3, 2nd day (B) = 2

2 days work = 5 units

60 ÷ 5 = 12 cycles

⏱ Time = 12 × 2 = 24 days (Ans)

10. STRANGE QUESTION (CATS & RATS)

(Q) 100 cats kill 100 rats in 100 days
How many days for 10 cats to kill 10 rats?

☞ Same ratio → SAME TIME

⏱ = 100 days (Ans)

PROFIT & LOSS

1. BASIC TERMS

CP = Cost Price
SP = Selling Price
MP = Marked Price

Profit = SP − CP → Profit
Loss = CP − SP → Loss

2. PROFIT & LOSS PERCENTAGE

$$\text{Profit \%} = \frac{\text{Profit}}{\text{CP}} \times 100$$

$$\text{Loss \%} = \frac{\text{Loss}}{\text{CP}} \times 100$$

$SP = CP \times \left(1 + \frac{P}{100}\right)$ → Profit

$SP = CP \times \left(1 - \frac{L}{100}\right)$ → Loss

3. WHEN SP & PROFIT % GIVEN

(Q) Profit = 20%, SP = 600

→ $CP = 600 \times \frac{100}{120} = \underline{\underline{500}}$ (Ans)

4. WHEN CP & LOSS % GIVEN

(Q) Loss = 25%, CP = 800

→ $SP = 800 \times \frac{75}{100} = \underline{\underline{600}}$ (Ans)

5. MARKED PRICE & DISCOUNT

Discount = MP − SP

$$\text{Discount \%} = \frac{\text{Discount}}{\text{MP}} \times 100$$

$$SP = MP \times \left(1 - \frac{d}{100}\right)$$

6. SUCCESSIVE DISCOUNTS

$$\text{Net discount \%} = a + b - \frac{ab}{100}$$

(Q) Discounts 20% & 10%

Net discount $= 20 + 10 - \frac{20 \times 10}{100}$

$= 30 - 2 = \underline{\underline{28\%}}$ (Ans)

7. PROFIT WITH DISCOUNT (IMPORTANT)

(Q) MP = 1000, Discount = 20%, Profit = 25%
Find CP.

→ $SP = 1000 \times 0.8 = 800$

→ $CP = 800 \times \frac{100}{125} = \underline{\underline{640}}$ (Ans)

8. FALSE WEIGHT PROBLE PIT

If a trader uses less weight:

$$\text{Profit \%} = \frac{\text{Weight difference}}{\text{Actual weight}} \times 100$$

(Q) Uses 900g instead of 1kg

$\text{Profit} = \frac{100}{900} \times 100 = \frac{100}{9} = 11\frac{1}{9}\%$ (Ans)

9. GAIN & LOSS ON SAME ITEM

(Q) Gain 10% on one item, Loss 10% on another. Net result?

Loss always

$$\text{Net loss \%} = \frac{a^2}{(100+a)^2} \times 100$$

For 10% → 0.83% loss

10. EQUAL SELLING PRICE CASE

(Q) Two articles sold at ₹100 each, Gain 20% on one, loss 20% on other.

$$\text{Net loss \%} = \frac{a^2}{100}$$

$= \frac{20^2}{100} = \frac{400}{100} = \underline{\underline{4\%\text{ loss}}}$ (Ans)

11. BUY x GET y FREE

$$\text{Effective discount \%} = \frac{y}{x+y} \times 100$$

(Q) Buy 2 get 1 free

→ $\text{Discount} = \frac{1}{2+1} \times 100 = 33\frac{1}{3}\%$ (Ans)

12. SELLING ABOVE & BELOW CP

(Q) Selling at 20% profit instead of 10%,
Extra profit = ₹100
Find CP.
Difference = 20% − 10% = 10%

→ $CP = 100 \times \frac{100}{10} = \underline{\underline{1000}}$ (Ans)

13. PARTNERSHIP LINK (IMPORTANT)

Profit ∝ Investment × Time

Ratio of profits = $I_1T_1 : I_2T_2$

SIMPLE INTEREST & COMPOUND INTEREST

1. BASIC TERMS

P = Principal

R = Rate of Interest (% per annum)

T = Time (years)

SI = Simple Interest

CI = Compound Interest

A = Amount

2. SIMPLE INTEREST (SI)

Formula

$$\boxed{SI = \frac{P \times R \times T}{100}} \qquad \boxed{\text{Amount} = P + SI}$$

Example

P = ₹1000, R = 10%, T = 2 years

$$SI = \frac{1000 \times 10 \times 2}{100} = \underline{\underline{200}}$$

$$\text{Amount} = 1000 + 200 = \underline{\underline{1200}}$$

3. SHORTCUT METHOD (SI)

Interest for 1 year = R% of P

Interest for T years = T × (R% of P)

4. WHEN AMOUNT GIVEN AT DIFFERENT TIMES (SI)

Amount after 3 yrs = ₹900

Amount after 6 yrs = ₹1200

Find P & R.

- Extra interest for 3 yrs = 1200 − 900 = $\underline{\underline{300}}$
- 1 year interest = $\frac{300}{3} = \underline{\underline{100}}$
- P = Amount (3 yrs) − Interest (3 yrs)
  = 900 − 300 = $\underline{\underline{600}}$
- $R = \frac{\text{Interest for 1 yr}}{P} \times 100 = \frac{100}{600} \times 100 = \frac{100}{6} = \underline{\underline{16\frac{2}{3}\%}}$

5. TIME FOR MONEY TO DOUBLE (SI)

$$\boxed{T = \frac{100}{R}}$$

6. COMPOUND INTEREST (CI)

Formula (Yearly)

$$\boxed{A = P\left(1 + \frac{R}{100}\right)^T} \qquad \boxed{CI = A - P}$$

Example

P = 1000, R = 10%, T = 2

$$A = 1000\left(1 + \frac{10}{100}\right)^2 = 1000\,(1.1)^2$$

$$= 1000 \times 1.21 = \underline{\underline{1210}}$$

$$CI = 1210 - 1000 = \underline{\underline{210}}$$

7. HALF-YEARLY / QUARTERLY COMPOUNDING

Half-Yearly	Quarterly
Rate = $\frac{R}{2}$	Rate = $\frac{R}{4}$
Time = 2T	Time = 4T
$\boxed{A = P\left(1 + \frac{R}{200}\right)^{2T}}$	$\boxed{A = P\left(1 + \frac{R}{400}\right)^{4T}}$

8. DIFFERENCE BETWEEN CI & SI (IMPORTANT)

For 2 years: $\boxed{CI - SI = P\left(\frac{R}{100}\right)^2}$

(Q) Difference = 25, R = 10%

$$P = \frac{\text{Difference} \times 100^2}{R^2} = \frac{25 \times 100 \times 100}{10 \times 10} = \underline{\underline{2500}}$$

9. WHEN MONEY BECOMES MULTIPLE TIMES (CI)

If money doubles:

$$2P = P\left(1 + \frac{R}{100}\right)^T$$

If money becomes 8 times:

$$8 = 2^3 \Rightarrow T = 3 \times (\text{doubling time})$$

10. COMPARISON: SI vs CI

- CI > SI (always, except T = 1 year)
- Difference increases with time
- CI grows exponentially

11. IMPORTANT SHORTCUTS

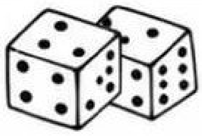
✓ For SI → Linear growth

✓ For CI → Exponential growth

✓ Use difference formula for 2-year questions

✓ Convert months → years carefully

PROBABILITY



1. BASIC DEFINITION

Probability of an event E :

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

2. SAMPLE SPACE

Sample space (S) → Set of all possible outcomes

(Q) Tossing a coin
→ $S = \{H, T\}$

(Q) Rolling a die
→ $S = \{1, 2, 3, 4, 5, 6\}$

3. RANGE OF PROBABILITY

$$0 \leq P(E) \leq 1$$

- $P(E) = 0$ → Impossible event
- $P(E) = 1$ → Certain event

4. COMPLEMENTARY EVENTS

If E is an event,

$$P(E') = 1 - P(E)$$

(Q) Probability of NOT getting a head
→ $= 1 - P(H)$

5. ADDITION THEOREM

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually Exclusive Events

If $A \cap B = \emptyset$,

$$P(A \cup B) = P(A) + P(B)$$

6. INDEPENDENT EVENTS

Two events A and B are independent if :

$$P(A \cap B) = P(A) \times P(B)$$

(Q) Tossing two coins
→ $P(HH) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

7. CONDITIONAL PROBABILITY

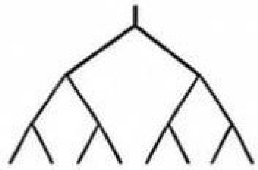
$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

(Q) Probability of A given B has occurred.

8. BAYES' THEOREM

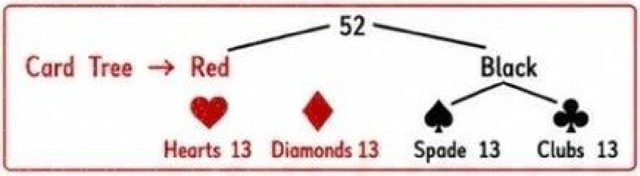
If $A_1, A_2, \ldots, A_n$ are mutually exclusive events and B is an event :

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{\sum P(A_j)P(B|A_j)}$$



⇒ Always draw tree diagram

9. PROBABILITY IN CARDS



- Face cards = 12, Kings = 4, Queens = 4, Aces = 4

(Q) Probability of a king → $\frac{4}{52} = \frac{1}{13}$

10. PROBABILITY IN DICE

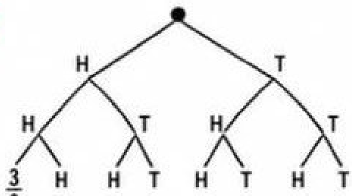
- 1 die → 6 outcomes
- 2 dice → 36 outcomes

(Q) Sum = 7
→ Favourable = 6 ; $P = \frac{6}{36} = \frac{1}{6}$

	1	2	3	4	5	6
1	2	3	4	5	6	6
2	2	3	4	5	6	7
3	3	4	5	6	7	8
4	4	5	6	7	8	9
5	5	6	7	8	9	10
6	7	8	9	10	11	12

11. PROBABILITY IN COINS

- 1 coin → 2 outcomes
- 2 coins → 4 outcomes
- 3 coins → 8 outcomes



(Q) Exactly 2 heads in 3 tosses = $\frac{3}{8}$

12. AT LEAST / AT MOST CONCEPT

"At least one" = 1 − (None)

(Q) At least one head in 2 tosses
→ $= 1 - P(TT) = 1 - \frac{1}{4} = \frac{3}{4}$

13. IMPORTANT SHORTCUTS

- ✓ Use complement for "at least"
- ✓ Independent → multiply
- ✓ Mutually exclusive → add
- ✓ Always write sample space

14. COMMON MISTAKES

- ✗ Forgetting total outcomes
- ✗ Confusing independent & exclusive
- ✗ Not subtracting intersection

RATIO & PROPORTION

1. BASIC DEFINITIONS

Ratio = Comparison of two quantities of same kind

$\boxed{a : b = \frac{a}{b}}$

Proportion :

$a : b :: c : d \Rightarrow \boxed{\frac{a}{b} = \frac{c}{d}}$

$\Rightarrow \boxed{ad = bc}$

2. PROPERTIES OF RATIO

- $a : b = ka : kb$
- $a : b = \frac{a}{k} : \frac{b}{k}$
- $a : b = b : a$ (inverse)

3. CONTINUED RATIO

$a : b = m : n$

$b : c = p : q$

$\Rightarrow a : b : c = mp : np : nq$

4. COMPOUND RATIO

$\boxed{(a : b) \times (c : d) = ac : bd}$

5. DIVIDING A QUANTITY IN A GIVEN RATIO

Sum = S , ratio = a : b

First part = $\boxed{S \times \frac{a}{a+b}}$

Second part = $\boxed{S \times \frac{b}{a+b}}$

(Q) Divide 300 in ratio 2 : 3

$\Rightarrow$ Total parts = 2+3 = 5

1 part = $\frac{300}{5} = 60$

Parts = $2\times60 = \underline{120}$ & $3\times60 = \underline{180}$

6. WHEN DIFFERENCE IS GIVEN

(Q) a : b = 3 : 5 , difference = 20

Let $a = 3x$, $b = 5x$

$5x - 3x = 20$

$2x = 20$

$x = 10$

$a = 3\times10 = \underline{30}$, $b = 5\times10 = \underline{50}$

7. MULTIPLE RATIOS

(Q) a : b = 2 : 3

b : c = 4 : 5 ?

LCM of b (3 & 4) = 12

$a : b = (2\times4) : (3\times4) = 8 : 12$

$b : c = (4\times3) : (5\times3) = 12 : 15$

$\Rightarrow a : b : c = \underline{8 : 12 : 15}$

8. WHEN SUM IS GIVEN (3 TERMS)

(Q) a : b : c = 3 : 4 : 5

Sum = 240

Total parts = 3+4+5 = 12

1 part = $\frac{240}{12} = 20 \Rightarrow a = 3\times20 = \underline{60}$

$b = 4\times20 = \underline{80}$

$c = 5\times20 = \underline{100}$

9. COMPARISON TYPE

(Q) If a : b = 3 : 2

Find $(4a + 5b) : (5a - 2b)$

Substitute a = 3, b = 2

$\Rightarrow (4\times3 + 5\times2) : (5\times3 - 2\times2)$

$\Rightarrow (12 + 10) : (15 - 4)$

$\Rightarrow 22 : 11 = \underline{2 : 1}$

10. PROPORTIONAL DIVISION

If $x \propto y \Rightarrow \boxed{\frac{x}{y} = \text{constant}}$

(Q) If x : y = 5 : 7 and x + y = 144

Total parts = 5+7 = 12

1 part = $\frac{144}{12} = 12 \Rightarrow x = 5\times12 = \underline{60}$

$y = 7\times12 = \underline{84}$

11. IMPORTANT IDENTITIES

If a : b = m : n

- $(a+b) : (a-b) = (m+n) : (m-n)$
- $(a^2-b^2) : (ab) = (m^2-n^2) : (mn)$

12. MIXED POWER RATIOS

(Q) If a : b = 2 : 3

Find $a^2 : b^2 = 2^2 : 3^2 = \underline{4 : 9}$

Find $\sqrt{a} : \sqrt{b} = \sqrt{2} : \sqrt{3}$

13. VARIATION

Direct Variation $\boxed{x \propto y \Rightarrow x = ky}$

Inverse Variation $\boxed{x \propto \frac{1}{y} \Rightarrow xy = k}$

14. IMPORTANT SHORTCUTS

- ✓ Always convert to same units
- ✓ Use substitution for complex expressions
- ✓ If powers differ → cannot compare directly
- ✓ LCM trick saves time in chain ratios

15. COMMON MISTAKES

- × Ignoring units
- × Wrong LCM in chain ratio
- × Mixing sum & difference logic
- × Wrong unit conversion in divisibility

NUMBER SYSTEM

1. TYPES OF NUMBERS

→ Natural Numbers (N) : 1, 2, 3, ⋯

→ Whole Numbers (W) : 0, 1, 2, 3, ⋯

→ Integers (Z) : ⋯ −3, −2, −1, 0, 1, 2, 3 ⋯

→ Rational Numbers (Q) : $\boxed{\frac{p}{q},\ q \neq 0}$

→ Irrational Numbers : $\sqrt{2}$, $\sqrt{3}$, π

→ Real Numbers (R) : Rational + Irrational

2. EVEN & ODD NUMBERS

Even = 2n
Odd = 2n + 1

- Even + Even = Even
- Odd + Odd = Even
- Even + Odd = Odd
- Odd × Odd = Odd
- Even × Any = Even

3. PRIME & COMPOSITE NUMBERS

Prime → Exactly 2 factors (1 & itself)

Composite → More than 2 factors

Smallest prime = 2

Only even prime = 2

4. FACTORS & MULTIPLES

Factor → Divides exactly

Multiple → Product of number

(Q) Number of factors of n =

If $\boxed{n = p^a q^b r^c}$

Total factors = $\boxed{[(a+1)(b+1)(c+1)]}$

5. HCF & LCM

HCF → Lowest power of common primes

LCM → Highest power of all primes

$\boxed{\text{HCF} \times \text{LCM} = \text{Product of two numbers}}$

(Q) If HCF = 12, LCM = 360

Numbers = 12x, 12y

$12x \times 12y = 12 \times 360$

→ $xy = 30$

6. DIVISIBILITY RULES

- 2 → Last digit even
- 3 → Sum of digits divisible by 3
- 4 → Last two digits divisible by 4
- 5 → Last digit 0 or 5
- 6 → Divisible by 2 & 3
- 8 → Last three digits divisible by 8
- 9 → Sum of digits divisible by 9
- 11 → (Sum of odd place − even place) divisible by 11

7. REMAINDER THEOREM

If n divided by d gives remainder r

$\boxed{n = dq + r}$

(Q) Find remainder when 7^{25} divided by 6

$7 \equiv 1 \pmod 6$

$\Rightarrow 7^{25} \equiv 1^{25} \pmod 6$

⇒ Remainder = 1 (Ans)

8. CYCLICITY (LAST DIGIT)

Last digits repeat in cycles of 4

2 → 2, 4, 8, 6

3 → 3, 9, 7, 1

7 → 7, 9, 3, 1

(Q) Find last digit of 7^{101}

101 mod 4 = 1

⇒ Last digit = $7^1 = 7$ (Ans)

10. FRACTIONS

Proper : numerator < denominator

Improper : numerator ≥ denominator

Convert mixed to improper

$$\boxed{a\frac{b}{c} = \frac{ac+b}{c}}$$

9. FINDING NUMBER USING HCF & LCM

(Q) Two numbers have HCF = 6

If one number = 30

Find other number

Other = $\frac{6 \times 180}{30} = 6 \times 6 = 36.$

12. SURDS

$\boxed{\sqrt{a} \times \sqrt{b} = \sqrt{ab}}$

$\boxed{\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}} = 5\sqrt{2}$

11. DECIMAL TO FRACTION

Let x = 0.333...

10x = 3.333...

9x = 3

$x = \frac{3}{9} = \frac{1}{3}$ (Ans)

13. SIMPLIFICATION TRICKS

- BODMAS rule
- Always simplify to fractions if needed

13. SIMPLIFICATION TRICKS

✓ Use prime factorization

✓ Cyclicity saves time

✓ Last digit saves time

IMPORTANT SHORTCUTS

✓ Use prime factorization

✓ HCF × LCM trick

✓ Last digit & remainder checks

15. COMMON MISTAKES

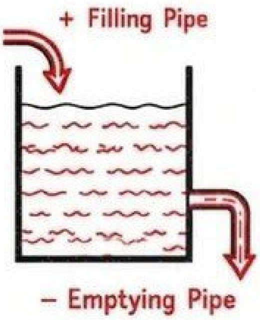
× Ignoring sign of integers

× Wrong cyclic power

× Mixing HCF & LCM rules

× Wrong conversion in divisibility

PIPES & CISTERNS



1. BASIC CONCEPT

Pipes & Cisterns is a Time & Work problem.

- → Tank Capacity = Work (W)
- → Filling pipe → Positive (+)
- → Emptying pipe → Negative (–)

$$\boxed{\text{Rate} = \frac{\text{Work}}{\text{Time}}}$$

2. UNIT METHOD (MOST IMPORTANT)

(Q) Pipe A fills a tank in 10 hrs,
Pipe B fills in 20 hrs.
Find time together.

LCM of 10 & 20 = $\boxed{\text{20 units (TC)}}$

A = 20/10 = 2 units/hr

B = 20/20 = 1 unit/hr

A + B = 3 units/hr

Time = $\frac{20}{3}$ hrs (Ans)

3. INLET & OUTLET TOGETHER

(Q) A fills tank in 10 hrs,
B empties tank in 40 hrs.
Find time together.

TC = LCM(10, 40) = $\boxed{\text{40 units (TC)}}$

A = +4 units/hr

B = –1 unit/hr

Net rate = 3 units/hr

Time = $\frac{40}{3}$ hrs (Ans)

4. LEAK IN THE TANK

(Q) A pipe fills tank in 12 hrs,
but due to a leak it fills in 15 hrs.
Find time to empty tank when full.

TC = LCM(12, 15) = $\boxed{\text{60 units (TC)}}$

Filling rate = $\frac{60}{12}$ = 5 units/hr

Net rate = 6/15 = 4 units/hr

Leak rate = 5 – 4 = 1 unit/hr

Emptying time = $\frac{60}{1}$ = 60 hrs (Ans)

5. ALTERNATE OPENING OF PIPES

(Q) Pipe A fills in 10 hrs,
Pipe B empties in 20 hrs.
A & B are opened alternately starting with A.
Find total time.

TC = LCM(10, 20) = $\boxed{\text{20 units (TC)}}$

A = +2 units/hr

B = –1 unit/hr

In 2 hrs → +1 unit

So, 20 units will take 40 cycles (20 × 2)

Time = 40 cycles = 80 hrs (Ans)

6. TWO PIPES FILLING, ONE EMPTYING

(Q) A fills in 10 hrs,
B fills in 15 hrs,
C empties in 30 hrs.

TC = LCM(10, 15, 30) = $\boxed{\text{30 units (TC)}}$

A = +3 units/hr

B = +2 units/hr

C = –1 unit/hr

Net rate = 3 + 2 – 1 = 4 units/hr

Time = $\frac{30}{4}$ = 7.5 hrs (Ans)

7. PARTIAL FILLING & EMPTYING

(Q) Tank is filled in 8 hrs,
but empties in 12 hrs when full.
Pipe is opened when tank is half full.
Find time to empty remaining.

TC = LCM(8, 12) = $\boxed{\text{24 units (TC)}}$

Half tank = 12 units

Emptying rate = $\frac{24}{12}$ = 2 units/hr

Time = $\frac{12}{2}$ = 6 hrs (Ans)

8. FRACTIONAL PART PROBLEMS

(Q) A pipe fills $^1/_3$ tank per hour,
another empties $^1/_6$ tank per hour.
Find net time.

Net rate = $\frac{1}{3} - \frac{1}{6} = \frac{2}{6} - \frac{1}{6} = \frac{1}{6}$ tank/hr

Time = 6 hrs (Ans)

9. SHORTCUT RULES

- If filling time > emptying time → tank fills
- If emptying rate > filling rate → tank never fills
- Always take LCM as Total Capacity
- Negative sign = emptying pipe

10. COMMON MISTAKES

- ✕ Forgetting negative sign
- ✕ Mixing time & rate
- ✕ Not using LCM
- ✕ Wrong unit conversion